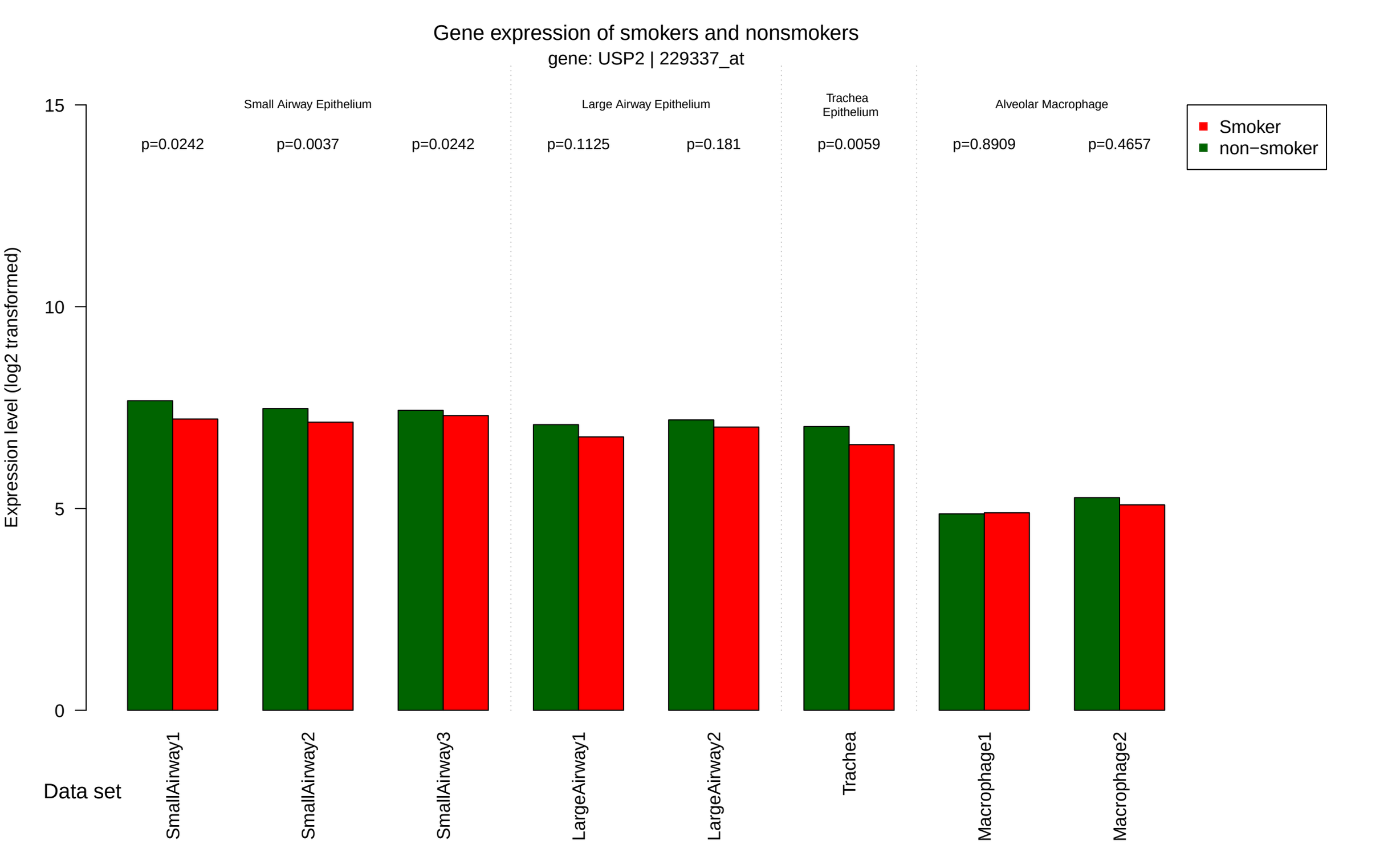
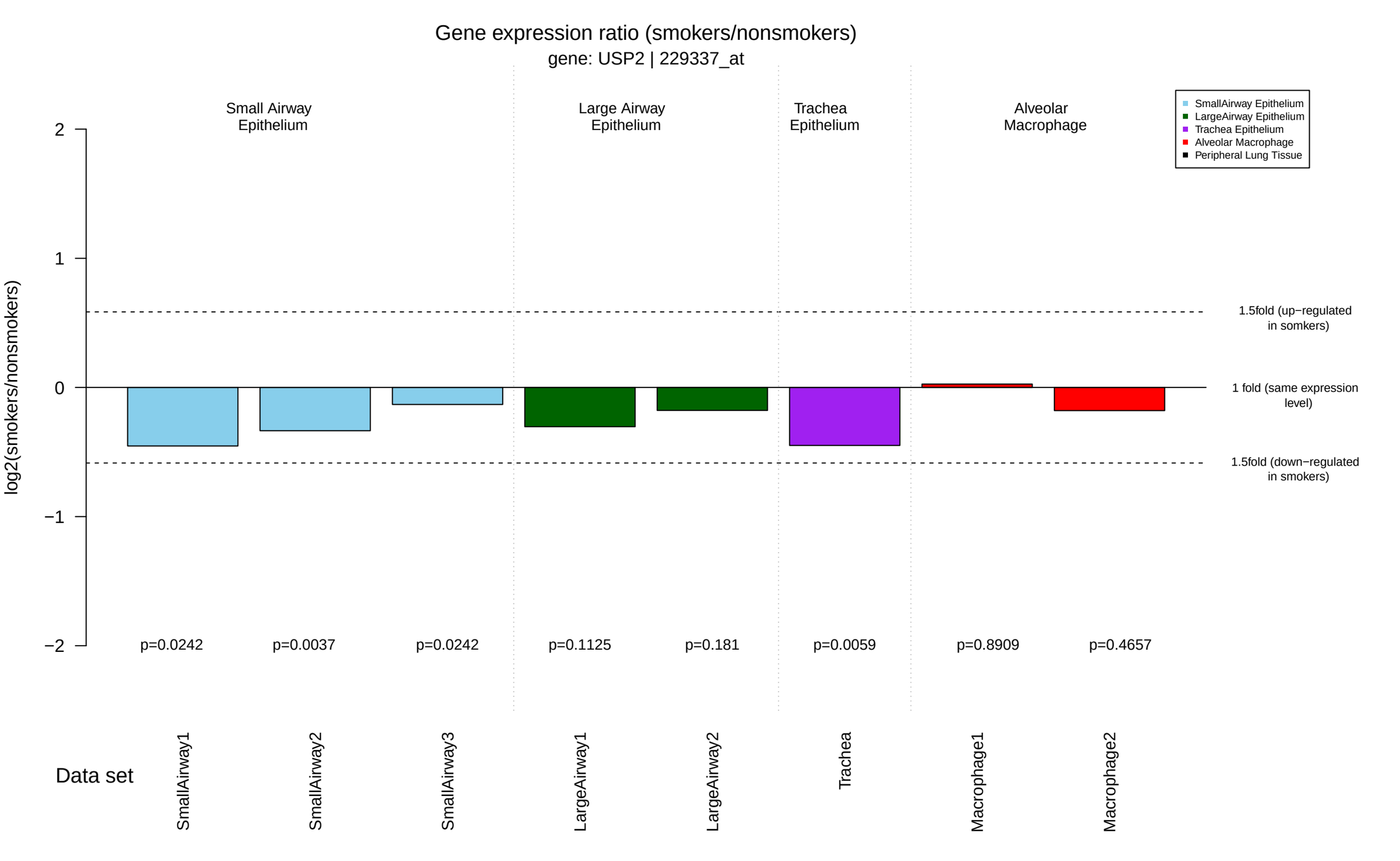
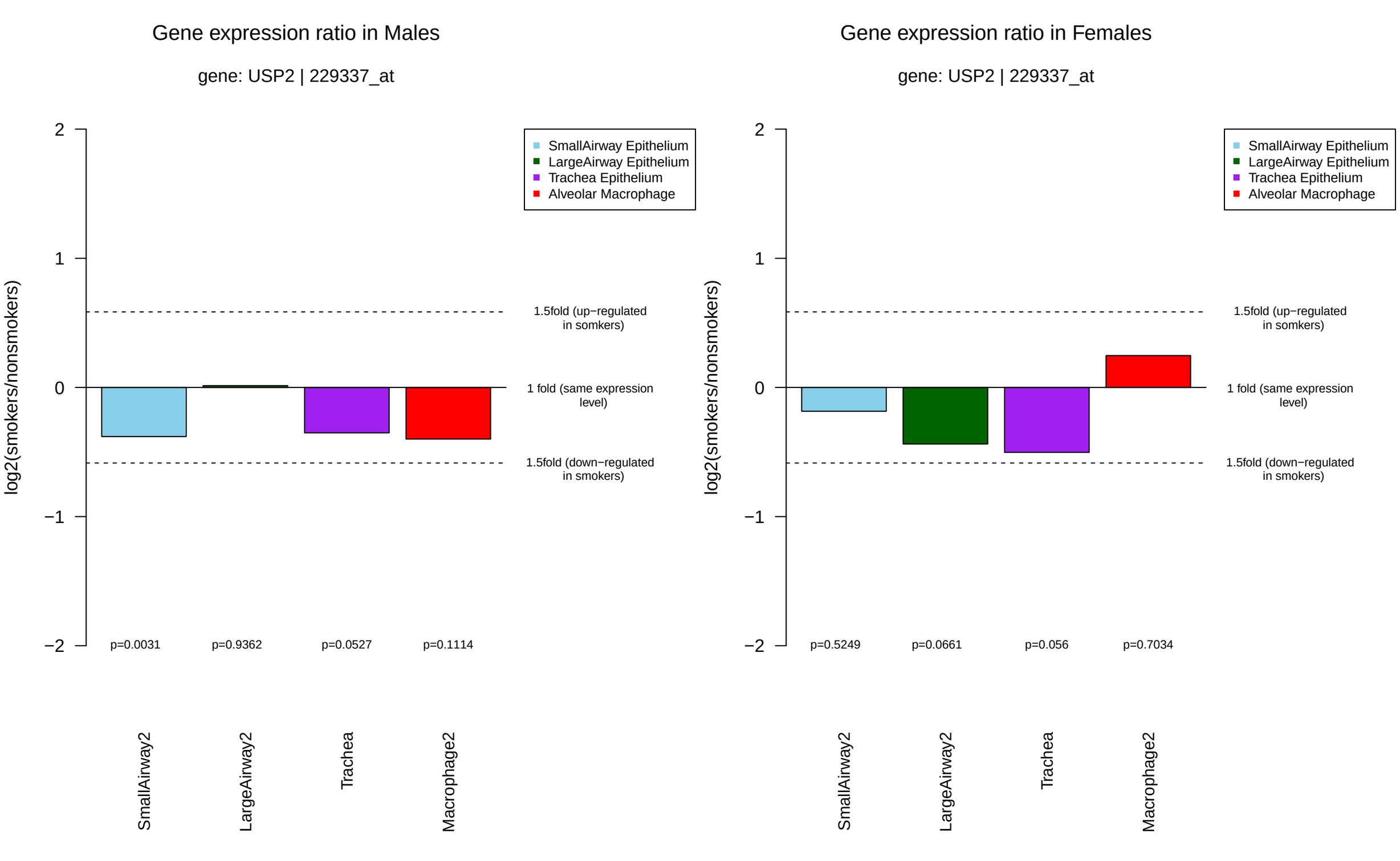
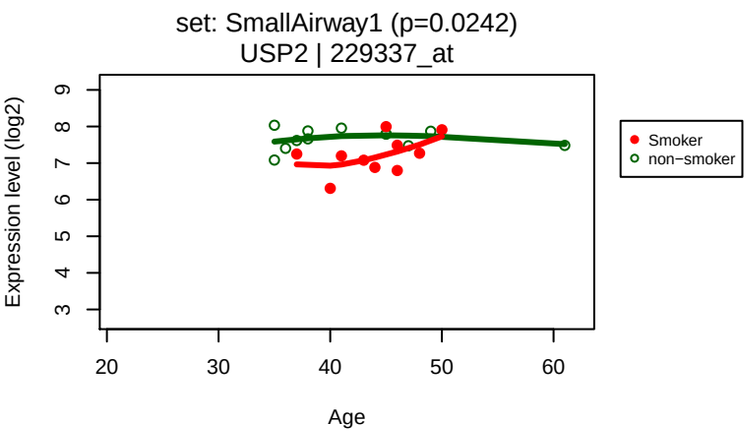
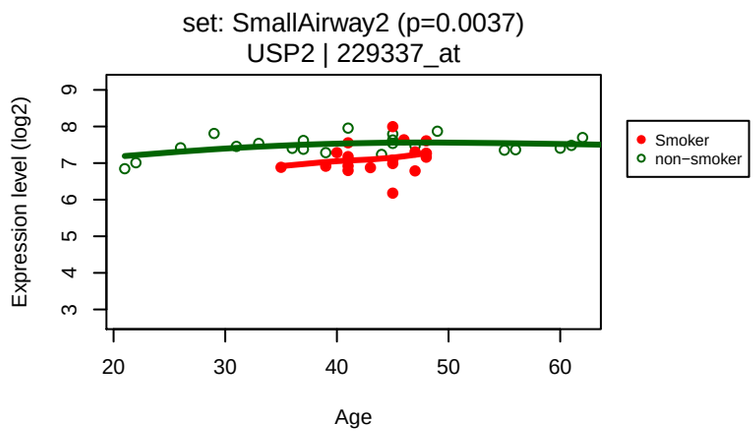
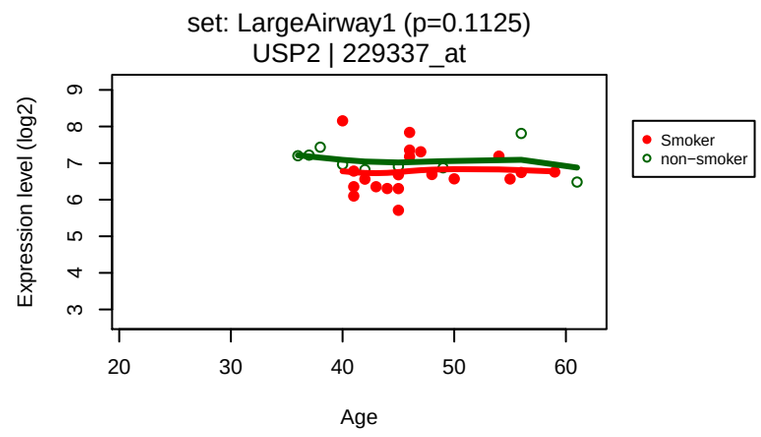

Gene expression of smokers and nonsmokers
gene: USP2 | 229337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0242
p=0.0037
p=0.0242
p=0.1125
p=0.181
p=0.0059
p=0.8909
p=0.4657
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

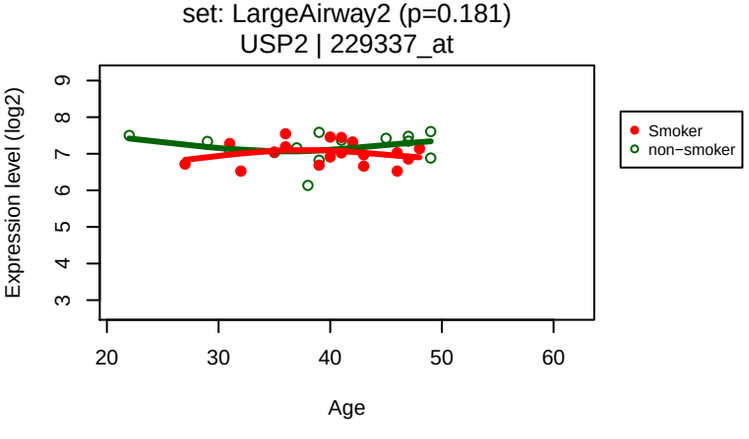

Gene expression ratio (smokers/nonsmokers)
gene: USP2 | 229337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0242
p=0.0037
p=0.0242
p=0.1125
p=0.181
p=0.0059
p=0.8909
p=0.4657
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

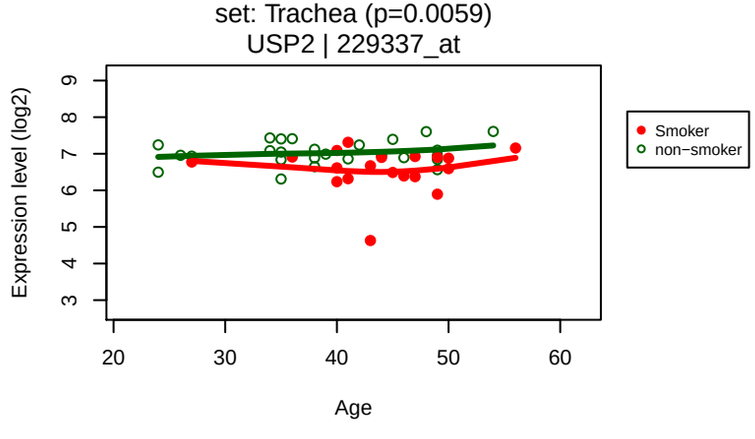

Gene expression ratio in Males
gene: USP2 | 229337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.9362
p=0.0527
p=0.1114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP2 | 229337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5249
p=0.0661
p=0.056
p=0.7034
SmallAirway2
LargeAirway2
Trachea
Macrophage2

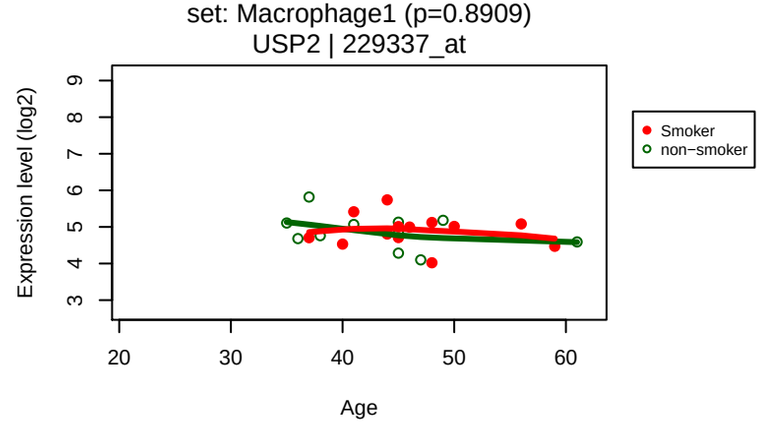

set: SmallAirway1 (p=0.0242)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker

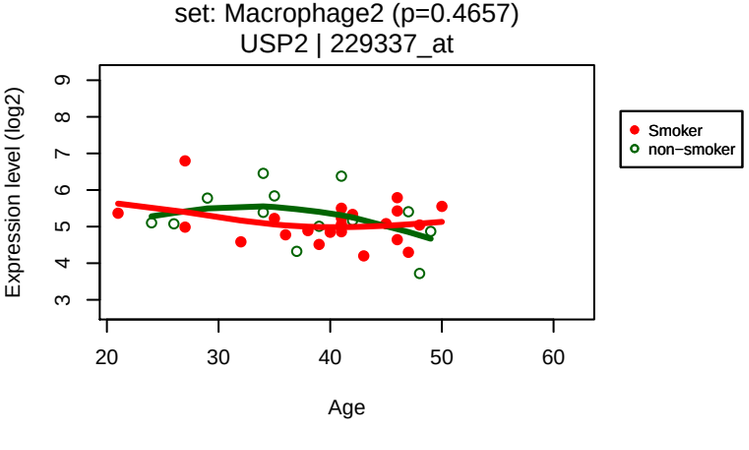

set: SmallAirway2 (p=0.0037)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker

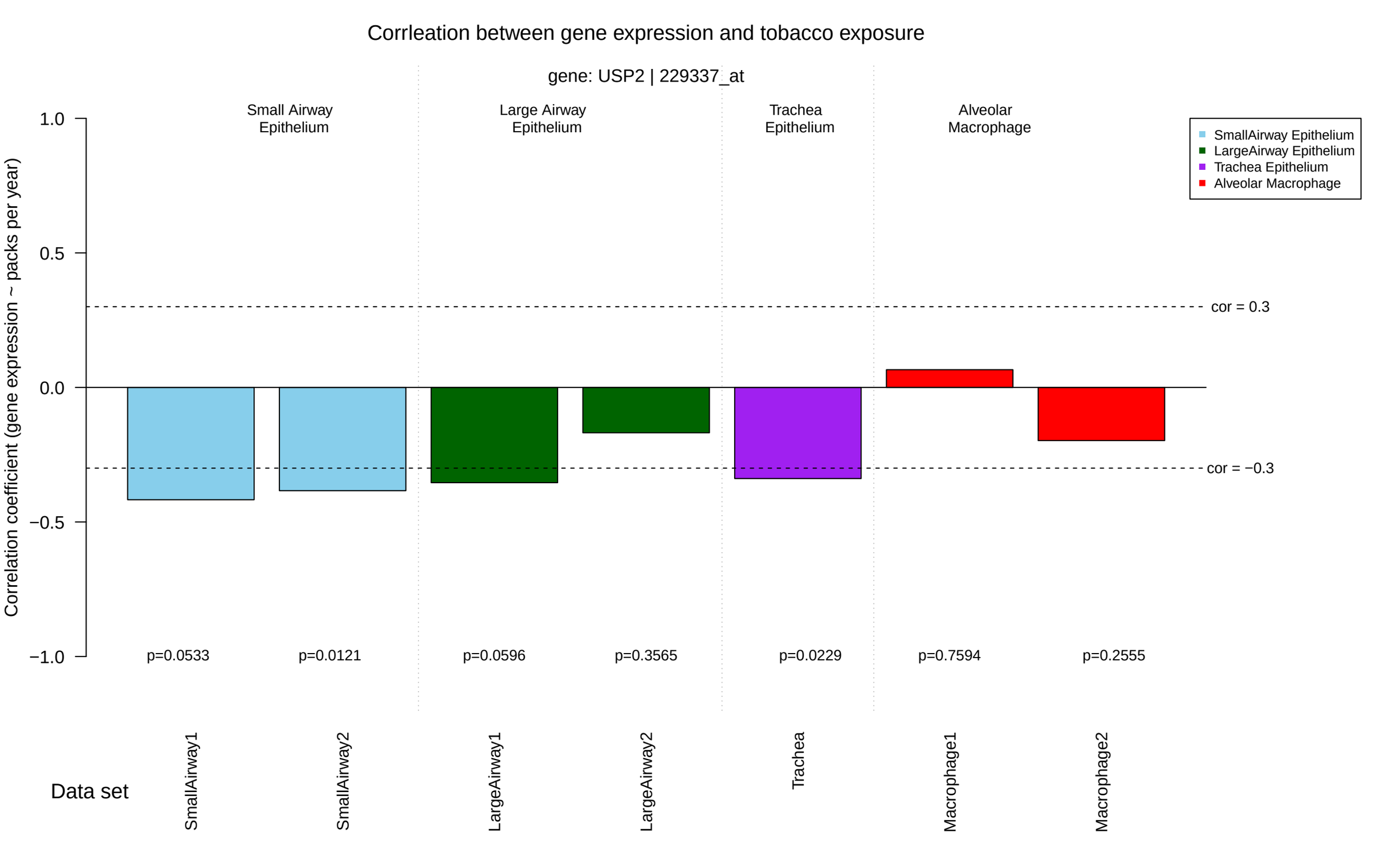

set: LargeAirway1 (p=0.1125)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.181)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0059)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8909)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4657)
USP2 | 229337_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP2 | 229337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0533
p=0.0121
p=0.0596
p=0.3565
p=0.0229
p=0.7594
p=0.2555
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2